Case 1: Gender shades: Intersectional accuracy disparities in commercial gender classification (Buolamwini & Gebru, 2018)

Bias identified and how it was measured	**Who was harmed and through what institutional mechanism**	**Root cause classification (historical, representation, measurement, aggregation, deployment) + 1 sentence for each type**	**Remediation proposed/implemented (Sufficient? Why/why not?)**
different ability of facial recognition AI to correctly identify people (particularly darker females), measured as a % correct from a dataset the researchers developed	no one directly since the paper was only about testing their new dataset; however, their literature review mentions potential ramifications in police work	Representation bias: The datasets that Buolamwini and Gebru (2018) were seeking to replace included a disproportionate number of lighter and male faces.	The article did not mention any remediation, though one of our reading questions mentions that what remediation was done, if any, was unsuccessful.

Case 2: Dissecting racial bias in an algorithm used to manage the health of populations (Obermeyer et al., 2019)

Bias identified and how it was measured	**Who was harmed and through what institutional mechanism**	**Root cause classification (historical, representation, measurement, aggregation, deployment)**	**Remediation proposed/implemented (Sufficient? Why/why not?)**
Black patients were being disproportionately excluded from high-risk care management programs due to a health care algorithm. The researchers	Black patients were harmed because they should have been enrolled into high-risk care management programs at a much higher rate than they were. This was because	historical: Due to sociocultural and historical factors, Black patients face more barriers in accessing and getting care, causing them to spend less, which	The authors proposed a solution which used combined health prediction and cost prediction to assign patients to care management programs,

demonstrated that Black patients at the same risk score tended to be sicker than White patients with the same score.	the way the algorithm was set up: even though it didn't explicitly include race, the fact that Black patients tended to spend less on health care meant that they got less care because cost was integrated into the algorithm.	led the algorithm to systematically underassign them to care management programs.	which they said resulted in an 84% decrease in bias. This is a large improvement, but whether it is sufficient depends on what level of bias, if any, one is willing to accept.

Synthesis

What do these two cases share as root causes, and what differs in the harm mechanism?

Both of these cases shared the root cause of the data that was used to train the algorithm. They differed in that the data in one example was focusing more or less explicitly on race, while the other dataset was (theoretically) colorblind. The harm mechanisms were different because of their different applications. In the Buolamwini and Gebru (2018) article, the theoretical harm is from use of facial recognition algorithms in, for example, law enforcement, which could result in someone being misidentified as the suspect for a crime. In the Obermeyer et al. (2019) article, the harm was that patients who should have been enrolled into a care management program were not enrolled due to indirect racial bias in the algorithm, which led to their getting less health care than they should have gotten.